

Ichthyology, Exam 1, F14**Name:** _____

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.
(2 points each)

Buccal Cavity	Iteroparity	Osmoregulator	Semelparity
Buccal Forcing	Labriform	Ostraciform	Sinus Venosus
Bulbous Arteriosus	Lowdown Dirtymean	Physostomous	Stenohaline
Carangiform	Nest Raiding	Pneumatic Duct	Subcarangiform
Conus Arteriosus	Opercular	Ram-Gill Ventilation	Tetraodontiform
Egg Raiding	Oscillation	Rectal Gland	Thunniform
Euryhaline	Osmoconformer	Salt Duct	Undulation
Force Ventilation	Osmohaline	Satellite	Ventricle

1. _____ Breathing mechanism used by strong swimmers at high speeds.
2. _____ Spawn once, and then die.
3. _____ Form of BCF swimming where the energy wave begins in the caudal penduncle.
4. _____ Chamber that collects deoxygenated blood from the body.
5. _____ Osmotic concentration of hagfish equals surrounding water.
6. _____ Tolerates a narrow range of salinity.
7. _____ Male steals nest of another male, guards his own and other male's eggs.
8. _____ Form of MPF swimming that uses oscillation of the pectoral fins.
9. _____ Chamber with the highest pressure during two-phase respiration.
10. _____ Used by sharks and rays to remove excess sodium ions.

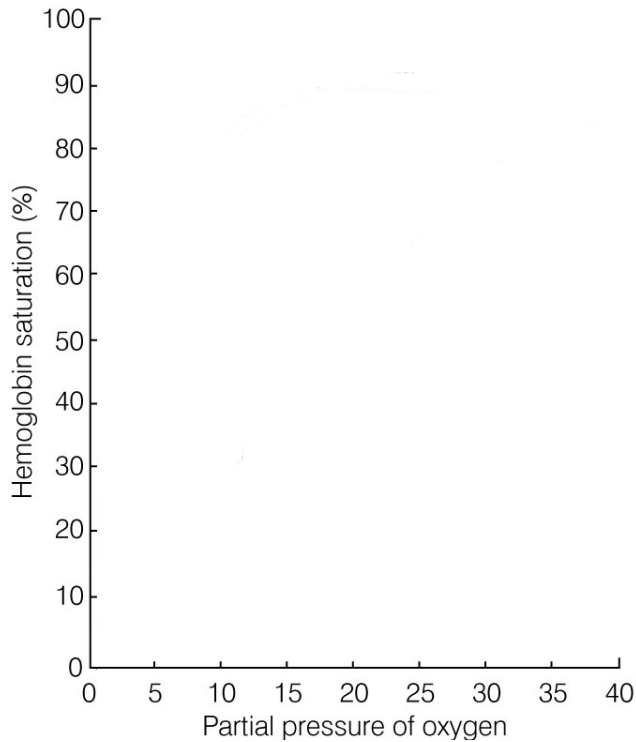
(GSH) Answer the next two questions. Terms are not in the word bank. (2.5 points each.)

11. _____ A modified gill structure used by some fishes for air breathing (two possible answers).
12. _____ Branchial arteries that leave the gills.

(5 pts) 13. List the five classes that together comprise “fishes.”

- a. _____ c. _____ c. _____
- d. _____ e. _____

(10 pts) 14. Draw an oxygen dissociation curve for a typical fast-swimming fish in a normoxic environment on the graph below. Draw a second dissociation curve for a typical sluggish fish in a hypoxic environment. Indicate P_{50} on the graph and two curves. Explain each curve along with your reasoning for the shape of the curve. *Be sure your drawings shows clear and accurate differences between the two fishes.*



(5 pts) 15. **(GSH)** If a typical fish has a relative gill surface area of 1000, would you expect the sluggish fish to have a similar, much higher, or much lower relative gill surface area. Explain why.

- (10 pts) 16. How do active fishes like tuna fishes maintain a core body temperature several degrees higher than their surrounding environment?
- (15 pts) 17. Sharks and tunas both cruise the open seas. Despite this commonality, these two fishes have very different body forms. Paul Webb, on pages 80–81 of his Scientific American article, “Form and Function in Fish Swimming,” explains why these two fishes differ so greatly. Describe one physiological, one structural and one ecological difference between sharks and tunas that explain the overall morphological difference in these cruising fishes.

- (20 pts) 18. Diagram and explain how a physoclistous bladder is inflated and emptied. Be sure to include all of the key structures and physiological processes. Indicate proper direction of flow for all molecules involved. Use representative values for pH and oxygen concentration to aid your explanation.

- (20 pts) 19. Explain how stenohaline freshwater and saltwater bony fishes maintain osmotic homeostasis. Explain for water and for relevant ions (e.g., Na^+ , Cl^- , K^+ , Ca^{2+}). Include for each when diffusion is operating and when active transport is required, and the primary regions for entry and exit of water and ions. Use illustrations to enhance your explanation. Do not include *elasmobranchs* or *hagfishes*.

- (10 pts) 20. **(GSH)** Sharks and rays use urea and TMAO to maintain osmotic balance. According to your text, another class of fishes also use urea and TMAO. Name this class. is the shared use of urea and TMAO the result of common ancestry or convergent evolution? Explain.
- (10 pts) 21. **(GSH)** O₂ has slightly lower solubility in saltwater than in freshwater but most air-breathing fishes are freshwater species. Provide a hypothesis with justification to explain why air-breathing evolved several times in freshwater fishes.